Fairness under Graph Uncertainty: Achieving Interventional Fairness with Partially Known Causal Graphs over Clusters of Variables

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Problem Setup & Motivation: Can we achieve fairness using a *cluster causal graph*?

Input Training data

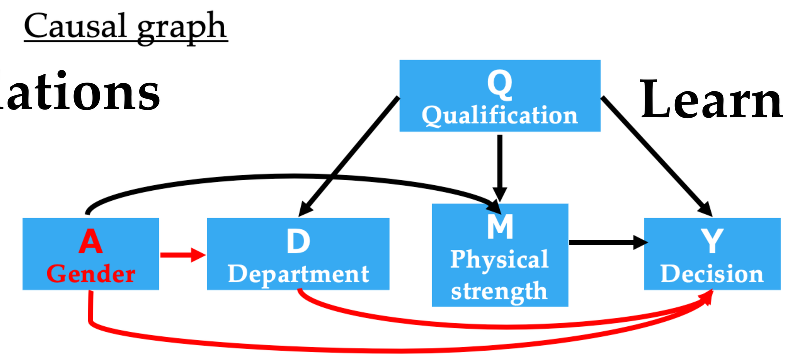
Sensitive feature

A Gender	Q Qualification	D Department	M Physical strength	Y Decision
Female	A	Economics	B	Accept
Male	B	Literature	B	Accept
Male	C	Science	C	Reject

$X = [A, Q, D, M]$

Infer causal relations

Causal graph



Learn an ML model

Output Fair and accurate ML model

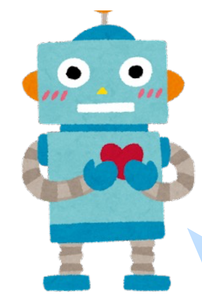
$$\hat{Y} = h_\theta(X)$$

Learn model parameters θ

$$\min_\theta \frac{1}{n}\sum_{i=1}^{n} \ell(y_i, h_\theta(x_i)) + \lambda g_\theta(x_1, \ldots, x_n),$$

Accurate causal graph inference is challenging in high-dimensional settings ☹

- We are NOT interested in the causal graph itself!
- **We just want to ensure fairness**

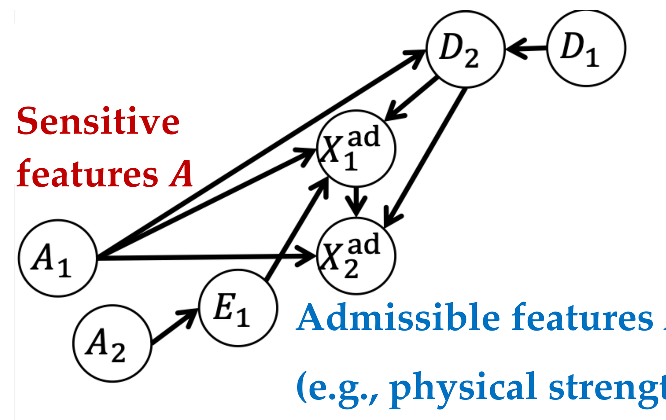


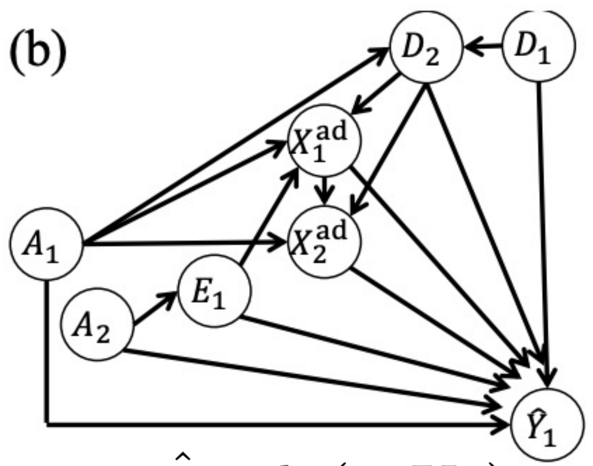
Q: Can we ensure fairness using a *high-level* causal graph structure representing macro-level relations among groups of variables?

Interventional Fairness [Salimi+; SIDMOD2019]

Measuring fairness via lens of causality

(a) Causal graph over features X (b) Causal graph over features X & prediction $\hat{Y}$





An SCM formalizes data-generating processes: $X_1^{\mathrm{ad}} = f_{X_1^{\mathrm{ad}}}(A_1, D_2, E_1, U_{X_1^{\mathrm{ad}}})$

Given by $\hat{Y} = h_\theta(X, U_{\hat{Y}})$

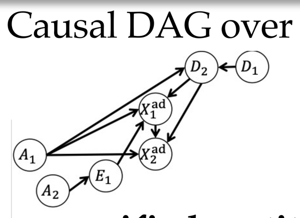
Definition Distributional equality under different *interventions*

$$P(\hat{Y} = \hat{y} \mid do(A = a), do(X^{\mathrm{ad}} = x^{\mathrm{ad}})) = P(\hat{Y} = \hat{y} \mid do(A = a'), do(X^{\mathrm{ad}} = x^{\mathrm{ad}}))$$

Cluster DAGs & Cluster CPDAGs

Cluster DAG Definition

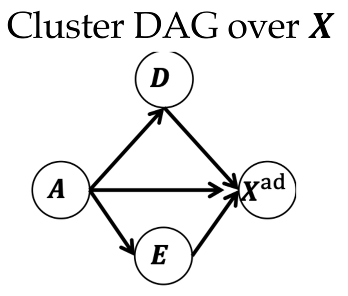
(a) Causal DAG over X



User-specified partition $\mathcal{C} = \{C_1, \ldots, C_d\}$

Assumption: No cycle is induced by partition

(c) Cluster DAG over X

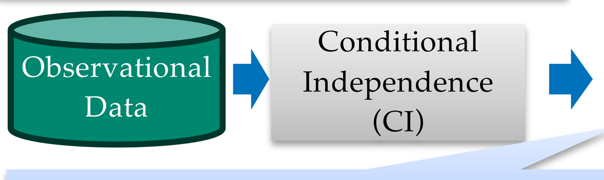


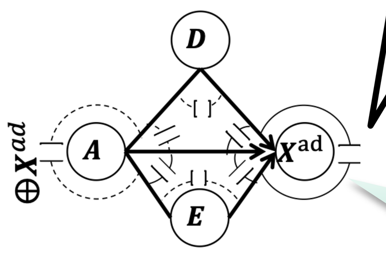
$C_i \to C_j$

if some $V_i \in C_i$ and $V_j \in C_j$ have an edge $V_i \to V_j$ in G_v

Graph Candidate Representation

(d) Cluster CPDAG over X

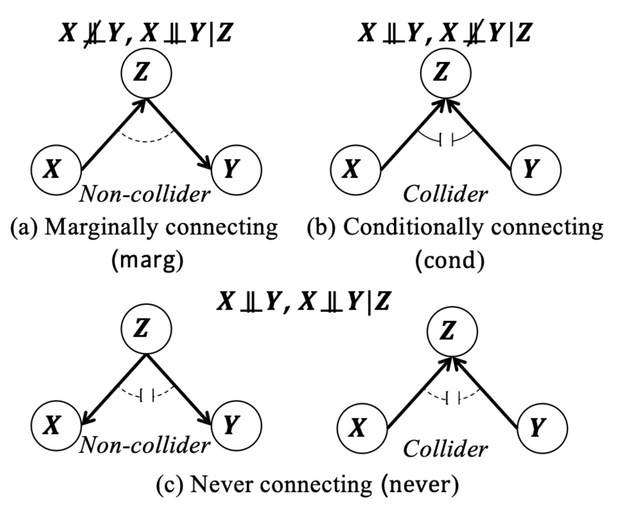




Represent a *cluster MEC*: a class of cluster DAGs with identical CI relations

Cluster-level edges alone are insufficient!

CI relations are expressed using the three states of *independence arcs*:



Easy to estimate ☺

Number of required CI tests is much smaller

Difficulty & Our Proposed Strategies

Q: How can we infer interventional distributions using a cluster CPDAG?

If we have access to a single variable-level DAG: Adjustment formula applies ☺

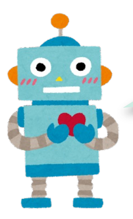
Identify back-door adjustment variables Z from the DAG and compute

$$P(\hat{Y} = \hat{y} \mid do(A = a), do(X^{\mathrm{ad}} = x^{\mathrm{ad}})) = \mathbb{E}_{Z, X^{\mathrm{re}} \mid a, x^{\mathrm{ad}}}\left[P(\hat{Y} = \hat{y} \mid A = a, X^{\mathrm{ad}} = x^{\mathrm{ad}}, Z, X^{\mathrm{re}})\right]$$

Difficulty: Cluster CPDAG represents **multiple DAGs** (in the cluster MEC) ☹

Our answer: Obtain *adjustment candidate sets* $Z^1, \ldots, Z^M$, at least one of which d-separates all back-door paths for the true DAG. We take 2 steps:

1. **Parent Enumeration**: From possible DAGs, enumerate the parents of A as *Definite & possible parents*
$$Z^m := \mathrm{pa}(A) \cup S^m \quad \text{for } m = 1, \ldots, M$$
2. **Adjustment Set Completion**: Add clusters that are needed to achieve cluster d-separation [Anand+; NeurIPS2025]



Since it is uncertain which of $Z^1, \ldots, Z^M$ is valid for the true DAG, we will take the **maximum** of distributional distances **over** $Z^1, \ldots, Z^M$ to formulate unfairness penalty g_θ

Graphical Algorithm & Learning Framework

1. Possible parent conditions

Q: When can the *siblings* be the *possible parents* of A (i.e., the parents of A in some cluster DAGs in the MEC)?

Theorem 3.5: *A set of clusters* $S \subseteq \mathrm{sib}(A, \mathcal{G}) := \{C \in \mathcal{C} : C - A$ *in* $\mathcal{G}\}$ *in a cluster CPDAG* $\mathcal{G}$ *is a possible parent set of* A *if and only if it satisfies the following two conditions:*

1. ***Collider-structure compatibility****: All distinct* $U, V \in S$ *satisfy either (i)* U *and* V *are adjacent, or (ii)* U *and* V *are not adjacent and* $\mathcal{A}_{\langle U, A, V\rangle} =$ never.
2. ***No back-path condition****: Let* $\mathcal{G}_\to$ *be the directed subgraph of* $\mathcal{G}$ *obtained by removing all undirected edges. For every* $U \in S$ *and every* $V \in \mathrm{sib}(A, \mathcal{G}) \setminus S$*, there is no directed path from* V *to* U *in* $\mathcal{G}_\to$.

$$Z^m := \mathrm{pa}(A) \cup S^m$$

If Z^m contains **a *connection-marked* cluster** $\oplus C_x \ldots$ (i.e., If conditioning on some $C_x \in Z^m$ yields (conditional) dependence between some cluster pair)

2. Cluster addition algorithm

Algorithm 1 Adjustment Cluster Completion $(Z^m, A, \mathcal{G})$

```
Require: Z^m, A, and cluster CPDAG G over X
 Z ← Z^m in Eq. (6); Processed ← ∅
 Z_⊕ ← Z ∩ {C_x in ⊕C_x in A_⟨P,Q,R⟩ for all triples in G}
 Queue ← {(P,Q) : P ∈ Z_⊕; Q adjacent to P; Q ≠ A}
 while Queue ≠ ∅ do
     pop (P,Q) from Queue
     if (P,Q) ∈ Processed then
         continue
     Processed ← Processed ∪ {(P,Q)}
     for each R adjacent to Q in G and R ≠ P do
         if A_⟨P,Q,R⟩ = marg then
             Z ← Z ∪ {Q}
         else if A_⟨P,Q,R⟩ = never then
             Z ← Z ∪ {Q}
             if Some P_x ∈ Z in ⊕C_x in A_⟨P,Q,R⟩ then
                 push (Q,R) into Queue
         else if A_⟨P,Q,R⟩ = cond then
             E ← {Q} ∪ PossDesc(G,Q); E ← Z ∩ E
             if E ≠ ∅ and Z has no cluster in ⊘C_x in
 A_⟨P,Q,R⟩ then
                 return Unidentifiable
 if FinalCert(Z,A,G)^1 returns Unidentifiable then
     return Unidentifiable
 return (Z, OK)
```

Adjustment cluster sets $Z^1, \ldots, Z^M$

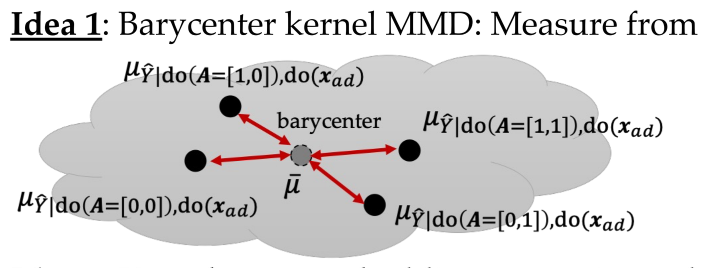
Worst-case unfairness penalization

Use the maximum of the kernel MMDs over $Z^1, \ldots, Z^M$:

$$\max_m \sum_{x^{\mathrm{ad}}} \sum_{a, a'} \mathrm{MMD}_m(P_{\hat{Y} \mid do(a), do(x^{\mathrm{ad}})}, P_{\hat{Y} \mid do(a'), do(x^{\mathrm{ad}})}),$$

which, however, needs $\mathcal{O}(M\, N_A^2\, N_{X^{\mathrm{ad}}}\, n^2)$ time ☹

Idea 1: Barycenter kernel MMD: Measure from the center



Idea 2: Kernel mean embedding approximated with RFFs

$$\mu_{\hat{Y} \mid do(a), do(x^{\mathrm{ad}})} := \mathbb{E}_{\hat{Y} \mid do(A=a), do(X^{\mathrm{ad}} = x^{\mathrm{ad}})}[\phi(\hat{Y})]$$

$$\phi_{\mathrm{RFF}}(y) = [\cos(\omega_1^\top y + b_1), \ldots, \cos(\omega_{d_{\mathrm{RFF}}}^\top y + b_{d_{\mathrm{RFF}}})]^\top$$

- *Random Fourier features* (RFFs) for Gaussian kernel [Rahimi+; NeurIPS2007]

Computationally efficient penalization

Penalty function with $\mathcal{O}(M N_A N_{X^{\mathrm{ad}}} n d_{\mathrm{RFF}})$ **time ☺**

$$g_\theta(x_1, \ldots, x_n) = \max_{m=1,\ldots,M} \sum_{x^{\mathrm{ad}}} \sum_{a} \|\hat{\mu}_{\hat{Y} \mid do(a), do(x^{\mathrm{ad}})} - \hat{\bar{\mu}}_{\hat{Y} \mid do(x^{\mathrm{ad}})}\|_2^2,$$

where kernel mean embedding is approximated as the RFFs using inverse probability weighting (IPW):

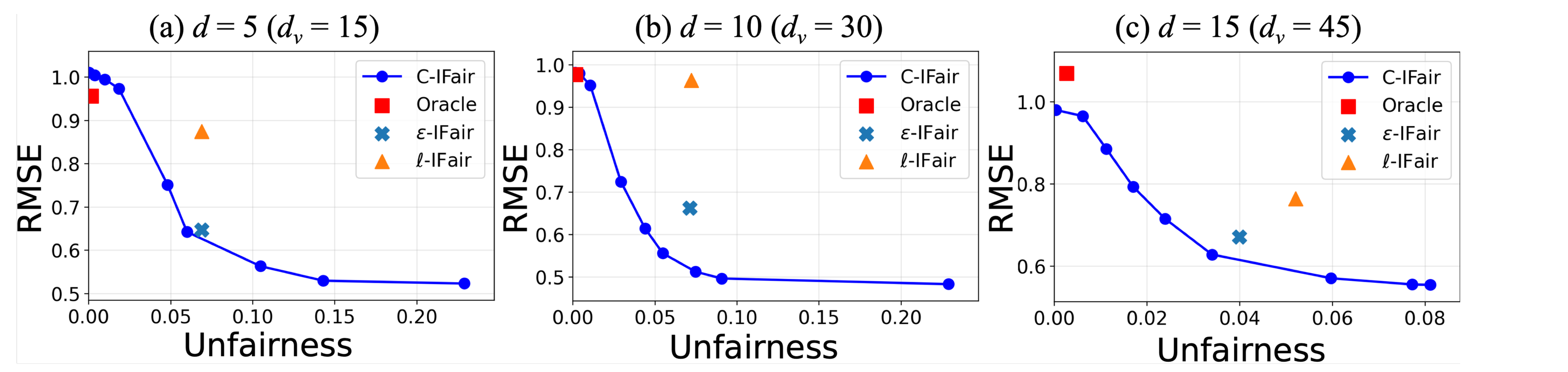
$$\hat{\mu}_{\hat{Y} \mid do(a), do(x^{\mathrm{ad}})} = \frac{1}{n}\sum_{i=1}^{n} w_i \phi_{\mathrm{RFF}}(\hat{y}_i)$$

IPW weight based on adjustment candidate Z^m:

$$w_i = \frac{\mathbf{I}(A_i = a, X_i^{\mathrm{ad}} = x^{\mathrm{ad}})}{\pi_{a, x^{\mathrm{ad}}}(z_i^m)}$$

Experimental Results

LINEAR	d=5 (d_v=15) RMSE↓	UNFAIRNESS↓	d=10 (d_v=30) RMSE↓	UNFAIRNESS↓	d=15 (d_v=45) RMSE↓	UNFAIRNESS↓
ORACLE	0.957 ± 0.037	0.000 ± 0.000	0.977 ± 0.044	0.000 ± 0.000	1.069 ± 0.143	0.000 ± 0.000
FULL	0.523 ± 0.195	0.259 ± 0.129	0.483 ± 0.195	0.229 ± 0.136	0.554 ± 0.210	0.081 ± 0.117
UNAWARE	0.774 ± 0.154	0.071 ± 0.059	0.774 ± 0.154	0.062 ± 0.079	0.793 ± 0.120	0.046 ± 0.104
NO-DESCS	0.739 ± 0.140	0.064 ± 0.089	0.739 ± 0.140	0.065 ± 0.100	0.699 ± 0.135	0.033 ± 0.028
ϵ-IFAIR	0.647 ± 0.030	0.069 ± 0.031	0.663 ± 0.062	0.071 ± 0.011	0.671 ± 0.044	0.040 ± 0.011
ℓ-IFAIR	0.875 ± 0.028	0.069 ± 0.081	0.964 ± 0.049	0.072 ± 0.052	0.764 ± 0.049	0.052 ± 0.052
C-IFAIR	0.643 ± 0.127	0.060 ± 0.054	0.660 ± 0.123	0.056 ± 0.052	0.669 ± 0.171	0.020 ± 0.036



	ADULT AUC↑	UNFAIRNESS↓	GERMAN AUC↑	UNFAIRNESS↓	OULAD AUC↑	UNFAIRNESS↓
ORACLE	0.709 ± 0.009	0.000 ± 0.000	0.582 ± 0.004	0.000 ± 0.000	0.628 ± 0.015	0.000 ± 0.000
FULL	0.874 ± 0.004	0.030 ± 0.024	0.762 ± 0.060	0.173 ± 0.011	0.697 ± 0.024	0.061 ± 0.001
UNAWARE	0.805 ± 0.049	0.018 ± 0.014	0.695 ± 0.061	0.101 ± 0.013	0.654 ± 0.019	0.004 ± 0.000
NO-DESCS	0.815 ± 0.007	0.026 ± 0.022	0.726 ± 0.059	0.085 ± 0.008	0.654 ± 0.020	0.004 ± 0.000
ϵ-IFAIR	0.812 ± 0.006	0.015 ± 0.012	0.756 ± 0.008	0.082 ± 0.000	0.641 ± 0.005	0.005 ± 0.002
ℓ-IFAIR	0.764 ± 0.002	0.015 ± 0.008	0.750 ± 0.035	0.151 ± 0.011	0.639 ± 0.011	0.003 ± 0.001
C-IFAIR	0.829 ± 0.021	0.014 ± 0.010	0.760 ± 0.061	0.065 ± 0.008	0.660 ± 0.018	0.001 ± 0.000